

Integer-Valued Entire Functions

Guy Sapire

10 June 2024

Based on chapter 9 of [Mas16].

1 Motivation

Something that we see is very helpful are functions that are entire and attain integer values at non-negative integer points, i.e $(f(n))_{n=0}^{\infty} \subseteq \mathbb{Z}$.

Example 1.

1. Any polynomial with integer coefficients.
2. Some polynomials with rational coefficients, for example $\binom{z}{k} = \frac{z(z-1)\dots(z-k+1)}{k!}$.
3. Exponential functions with base in $\mathbb{N}$, for example 2^z .
4. Some trigonometric functions, for example $f(z) = \sin(\pi z)$. Note that this function also has exponential growth in $\mathbb{C}$, because $\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}$. In fact, f grows more rapidly than 2^z , there are arbitrarily large z such that $|f(z)| \geq e^{\pi|z|}/3$.

Looking at these examples, We see that it is hard to think of examples with growth slower than 2^z that aren't polynomials. Pólya proves that there are no such functions.

2 Pólya

Theorem 2 (Pólya). *If f is an entire function such that $(f(n))_{n=0}^{\infty}$ are integers and $|f(z)| < cC^{|z|}$ for some $c > 0$ and $C < 2$ then f is a polynomial.*

Definition 3. Given an entire function f , we can formally expand it by its Newton interpolation $\sum_{k=0}^{\infty} a_k \binom{z}{k}$, where the coefficients are defined by

$$a_k = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(j)$$

Claim 4. *If $|f(z)| < cC^{|z|}$ for some $c > 0$ and $C < 2$, then the Newton interpolation converges to f .*

Proof. For any $n \in \mathbb{N}$ we can prove inductively that

$$\frac{1}{w-z} = \sum_{k=0}^{n-1} \frac{k!}{w(w-1)\dots(w-k)} \binom{z}{k} + \frac{n!}{w(w-1)\dots(w-n+1)(w-z)} \binom{z}{n}$$

Now multiplying both sides by $\frac{f(w)}{2\pi i}$ and integrating with regards to w on the circle $\{|w| = 2n\}$ for any n such that $2n \geq |z| + 1$, We get from Cauchy's theorem

$$f(z) = \sum_{k=0}^{n-1} \tilde{a}_k \binom{z}{k} + a_n(z) \binom{z}{n}$$

where

$$\tilde{a}_k = \frac{k!}{2\pi i} \int_{|w|=2n} \frac{f(w) dw}{w(w-1)\dots(w-k)} \stackrel{(*)}{=} a_k$$

(*) - From partial fraction decomposition. And

$$a_n(z) = \frac{n!}{2\pi i} \int_{|w|=2n} \frac{f(w) dw}{w(w-1)\dots(w-n+1)(w-z)}$$

Can be bounded

$$|a_n(z)| \stackrel{(*)}{\leq} \frac{2cnC^{2n}}{\binom{2n}{n}} \leq 2cn(2n+1)(C/2)^{2n} \xrightarrow{n \rightarrow \infty} 0$$

Where (*) follows from ML-inequality and the limit is 0 because $C/2 < 1$. Furthermore, denoting $m = \lfloor |z| \rfloor$, we have

$$\left| \binom{z}{n} \right| \leq \binom{m+n}{n} = \binom{m+n}{m} \leq (|z| + n)^m$$

So $a_n(z) \binom{z}{n}$ still converges to 0 as $n \rightarrow \infty$. □

Corollary 5. *From the formula for a_k that we got in the previous proof, we get bounds:*

$$|a_k| = \left| \frac{k!}{2\pi i} \int_{|w|=2k} \frac{f(w) dw}{w(w-1)\dots(w-k)} \right| \leq 2ck(2k+1)(C/2)^{2k}$$

Claim 6. *If $(f(n))_{n=0}^\infty \subseteq \mathbb{Z}$ then the coefficients a_k are integers.*

Proof. The coefficients are $\mathbb{Z}$ -linear combinations of $(f(n))_{n=0}^\infty$ and so they are in $\mathbb{Z}$. □

proof of Pólya. From the previous claims, we know that $f(z) = \sum_{k=0}^\infty a_k \binom{z}{k}$ where a_k are integers and for large enough k we have $|a_k| < 1$ so f is a polynomial. □

Example 7. It is fruitful to look at the example of 2^z again. Its Newton Interpolation is $\sum_{k=0}^\infty \binom{z}{k}$, because for this function the coefficients are

$$a_k = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} 2^j = (2-1)^k = 1$$

This shows that the condition $C < 2$ in the claim is tight: plugging in $z = -1$ the expansion is $1 - 1 + 1 - 1 + \dots$ which is not 2^{-1} (although close). Also note that $|a_k|$ aren't converging to 0, for the same reasons.

3 Waldschmidt

Waldschmidt noted that a weaker analog of Pólya's theorem, replacing the condition $C < 2$ with $C < C_0$ for some $C_0 < 2$, could be proved by means of an auxiliary polynomial. It is instructive to show such a result because it could be generalized, and inspires ideas to prove different theorems. For example, in a similar way Escassut and Mignotte proved the theorem for $C_0 = e^{1/283} = 1.003\dots$. We will sketch a proof right now, without working hard to get a good constant.

Assume f is an entire function such that $(f(n))_{n=0}^\infty \subseteq \mathbb{Z}$ and $|f(z)| < ce^{|z|/\lambda}$ for some λ (So $C_0 = e^{1/\lambda}$) and $c \geq 1$. We will keep adding constraints on λ as needed.

We don't know anything about the higher derivatives of f at any point, so instead of constructing an auxiliary polynomial with a zero of high degree, our best bet is to construct an auxiliary polynomial $\varphi(z) = P(z, f(z))$ with many zeros, for example at $0, 1, \dots, S-1$ for some $S \in \mathbb{N}$ and then force $\varphi(S)$ to be an integer small enough so it'll have to be 0. Thus we attained another zero and could continue by induction to get as many zeros as we want.

Take $M \in \mathbb{N}$, $L \in \mathbb{N}$ odd, and $p_{ij} \in \mathbb{Z}$ not all zero such that

$$\varphi(z) = \sum_{i=0}^L \sum_{j=0}^M p_{ij} \binom{z}{i} f(z)^j$$

satisfies $\varphi(0) = \varphi(1) = \dots = \varphi(S-1) = 0$ for some $S \in \mathbb{N}$. By Siegel's lemma we can take $S = \frac{(L+1)(M+1)}{2}$.

$\varphi(S)$ is an integer, now we can try to bound it to see that it is 0. Using Schwarz lemma and careful bounds (Stirling's approximation, etc) we get that for large enough M there is some constant A such that

$$|\varphi(S)| \leq A(L+1)^2 2^{-L} e^{3720L/\lambda}$$

And so for $\lambda > 3720/\log(2)$, for example $\lambda = 5367$, this bound converges to 0 as $L \rightarrow \infty$ and we can choose L large enough to get $\varphi(S) = 0$. This concludes the base of induction. The induction step involves a very similar bound, but a bit more difficult, which we skip.

After all of this, we see that $\varphi(n) = 0$ for any n and so by definition φ 's Newton expansion vanishes identically and could be shown that it converges to φ , so φ vanishes identically.

Going back to the beginning, this implies that f is algebraic over $C(z)$, meaning it is a polynomial. We finished the proof, with constant $C_0 = e^{1/5367} = 1.00018\dots$

In such a way you can extend the theorem to similar families of entire functions, for example functions that send $\mathbb{Z}[i]$ to itself.

Lemma 8 (Schwarz lemma for φ). *If φ has roots at $0, 1, \dots, k-1$ then we can write*

$$\tilde{\varphi}(z) := \frac{\varphi(z)}{z(z-1)\dots(z-(k-1))}$$

By the maximum modulus principle we have

$$|\tilde{\varphi}(k)| \leq \sup_{|z|=k} |\tilde{\varphi}(z)| \leq \sup_{|z|=3k} |\tilde{\varphi}(z)|$$

And note that

$$\begin{aligned} |\tilde{\varphi}(k)| &\geq k^{-k} |\varphi(k)| \\ \sup_{|z|=3k} |\tilde{\varphi}(z)| &\leq (2k)^{-k} \sup_{|z|=3k} |\varphi(k)| \end{aligned}$$

So we get $|\varphi(k)| \leq 2^{-k} \sup_{|z|=3k} |\varphi(k)|$

4 Further Work

further work was done, for example by using other means of interpolation. For example:

Theorem 9 (Selberg, 1941). *If f is an entire function such that $f(\mathbb{Z}_{\geq 0}) \subseteq \mathbb{Z}$ and $|f(z)| \leq cC^{|z|}$ for $c > 0$ and $C < 2e^{1/1500}$, then f is of the form $P_0(z) + P_1(z)2^z$ for some polynomials $P_0, P_1 \in \mathbb{Z}[x]$*

References

- [Mas16] David Masser. *Auxiliary Polynomials in Number Theory*. Cambridge Tracts in Mathematics. Cambridge University Press, 2016.